

MACHINE LEARNING & DATA SCIENCE CLUB

Intro to Python

No coding experience required

Beginner-friendly



```
# your first program  
print("Hello, MLDS!")
```

```
# output  
Hello, MLDS!
```

What is Python?

A high-level programming language: you write instructions in something close to plain English, not machine code.

1991

First released on February 20, 1991 by Guido van Rossum

1

General-purpose

Not tied to one kind of task or industry: the same language runs websites, analyses data and trains models.

2

Open-source

Free to use, modify and share, which is why so many people build tools for it.

Why do we use Python?

Easy to read

Clean syntax that looks close to English, so beginners can focus on ideas, not punctuation.

Runs anywhere

The same code works on Windows, Mac and Linux without major changes.

Huge community

Millions of users means answers, tutorials and ready-made libraries for almost anything.

The language of ML

Most machine learning, natural language processing, and data science work happens in Python, from research labs to industry.

The ML & data toolkit

Free libraries mean you almost never start from scratch. A few you'll run into:

WORK WITH DATA

```
import numpy as np
import pandas as pd
```

Fast math on arrays; tables like a spreadsheet

VISUALIZE

```
import matplotlib
import seaborn
```

Charts and plots from a few lines

CLASSIC ML

```
import sklearn
```

scikit-learn: regression, classification, clustering

DEEP LEARNING

```
import torch
import tensorflow
```

PyTorch & TensorFlow: build and train neural nets

LLMS & NLP

```
import transformers
```

Hugging Face: thousands of pretrained models

WHERE YOU CODE

```
Jupyter notebooks
```

Run code cell by cell, like today

Let's practice Python!

Open the notebook and follow along. Today's topics:

Data types

Variables

Print & input

Type conversion

Arithmetic

Conditionals

For loops

Functions

Lists

Dictionaries